

Size statistics of cumulus cloud populations in large-eddy simulations

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How many clouds of each size are there, and which sizes carry the cloud cover and the vertical mass flux?



Neggers, Jonker and Siebesma use large-eddy simulation (LES) of shallow cumulus to measure how many clouds of each size occur, and which sizes contribute most to the projected cloud cover and to the vertical mass flux. The motivation is parameterization in general circulation models: radiation schemes need the geometry of broken cloud fields, and mass-flux convection schemes, such as Arakawa and Schubert (1974), need the vertical transport by clouds. It is a validation study: the simulated clouds are analysed with exactly the algorithm used on satellite and aircraft images, with a comparable number of clouds, so the simulated statistics can be compared with observed ones. The dots on the cover are decorative.

Aims

- 1 Test LES cloud populations against observed ones, with the algorithm of satellite studies.
- 2 Find a universal form of the cloud size density.
- 3 Find which sizes dominate cover and mass flux.

Simulations: KNMI LES

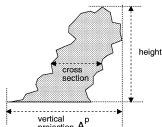
$\Delta x = \Delta y = 50$ m, $\Delta z = 40$ m, 6.4 km domain;
3D fields sampled every 5 min.

BOMEX	ocean, trade-wind cumulus, steady	36,776 clouds
SCMS	Florida, land, deepening layer	20,268 clouds
ARM SGP	Oklahoma, land, diurnal cycle	35,137 clouds

LES clouds are measured exactly as satellites measure real ones

Cloud statistics

Cloud n : connected saturated grid boxes, sized by the projected area A_n^p (right).



$$\ell_n = \sqrt{A_n^p}, \quad \mathcal{N}(l) = a l^b$$
$$\alpha^p(l) = \frac{l^2 \mathcal{N}(l)}{L_x L_y}, \quad \mu(l) = \overline{\alpha(l, z) w(l, z)}$$

$\mathcal{N}$: clouds per unit size. α^p : their share of the projected cover. μ : their share of the mass flux.

Question, simulations and cloud statistics

Three aims: validate LES cloud populations against observed ones, look for a universal functional form of the cloud size density, and find which cloud sizes dominate the cloud cover and the mass flux.

The KNMI LES solves filtered equations with a prognostic subgrid kinetic-energy scheme. The three cases come from field campaigns. BOMEX is steady trade-wind cumulus over the ocean, run ten times with randomized initial temperature profiles; the first three hours are discarded as spin-up. SCMS (Florida, 5 August 1995) has much stronger surface fluxes over land and a deepening cloud layer. ARM (Southern Great Plains, 21 June 1997) is a full diurnal cycle over land. Each reference case samples $2\text{--}4 \times 10^4$ clouds, comparable to the $\sim 10^4$ of satellite studies.

A cloud is a connected set of saturated grid boxes. Its linear size is the square root of its vertically projected area, the only size a satellite image can see. $\mathcal{N}(l)$ counts clouds per unit size and integrates to the total number N ; $\alpha^p(l)$ weights them by their projected area and integrates to the projected cloud cover a^p ; $\mu(l)$ is the mass flux carried by clouds of size l , the product of cloud fraction and cloud-mean vertical velocity averaged over the depth of the cloud layer, and it integrates to the total mass flux. Each cloud keeps its projected size at every height, so it stays in one bin. The fit is a power law, $\mathcal{N} = a l^b$. Histograms are shown in logarithmic bins, $\mathcal{N}^* = l \ln 10 \mathcal{N}$, whose log-log slope is $b + 1$.

A power law with a scale break, collapsed by one length

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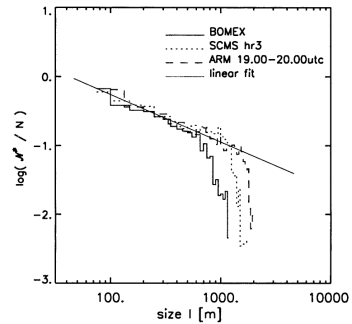


Fig. 4a: N^*/N for the three cases; fit $1.121 - 0.70 \log l$ below the break.

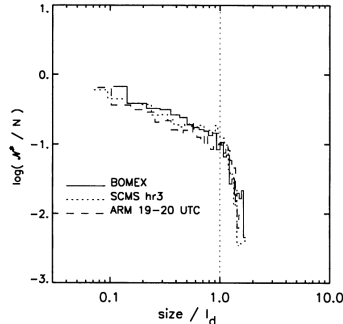


Fig. 6: the same densities against l/l_d ; dotted line at the break.

Below the break

$b = -1.70$ over about one decade, from the grid spacing to ℓ_d .

Observed: mean $b = -1.98$ (Benner & Curry 1998).

Scale break ℓ_d

BOMEX	700 m
SCMS	1,050 m
ARM, 15–16 UTC	400 m
ARM, 19–20 UTC	1,250 m

More sampled clouds: the same break.

One exponent fits every case; only the break ℓ_d changes

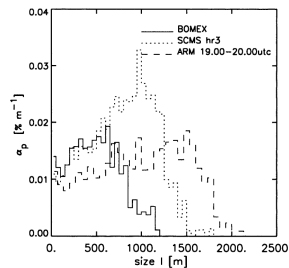
A power law with a scale break, collapsed by one length

Figure 4a shows the log-binned size density, normalised by the number of clouds. For sizes between the grid spacing and a break, all three cases collapse on one straight line of slope -0.70 , that is $b = -1.70$. Lognormal and exponential forms would curve on these axes, so the power law is preferred in this range. The exponent is close to observed values: Cahalan and Joseph report a log-bin slope of -0.89 for remotely sensed clouds, and Benner and Curry an average b of -1.98 for tropical shallow cumulus, with spread.

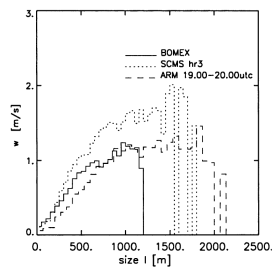
Above a size ℓ_d the density drops much faster. The break is not a sampling artefact: a convergence test with more clouds changes neither its position nor the histogram above it. Its value depends on the case (Table 2), and in ARM it moves from 400 m at 15–16 UTC to 1250 m at 19–20 UTC, through 700, 1000 and 1100 m in the intervening hours.

Figure 6 divides the size by ℓ_d : the three cases then collapse over all sizes, including above the break, where another exponent or another functional form applies. So ℓ_d is the only case-dependent parameter, which is what makes a universal description plausible. What controls it remains open. The authors mention the depth of the (sub)cloud layer, suggested by the ARM diurnal cycle, the largest convective cells of the boundary layer (Cahalan and Joseph), and humidity fluctuations in the subcloud layer, whose filtering changed the typical cloud size dramatically in Jonker et al. (1999).

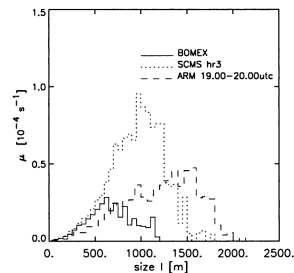
Intermediate clouds carry the cover and the mass flux



(a) projected cover $\alpha^p(l)$



(b) cloud-mean $w(l)$



(c) mass flux $\mu(l)$

Pure power law: $\alpha^p \propto l^{b+2}$, so the smallest clouds would dominate if $b < -2$, the largest if $b > -2$. Here $b = -1.70$ below ℓ_d and $b < -2$ above: α^p peaks at the break.

$\mu = \alpha w$ peaks at larger sizes still, since w grows with l . The smallest clouds carry almost no mass flux.

Robustness

Grid 25–100 m:
same dominant and largest sizes.

Domain 3.2 km:
the largest clouds are missing.

Shear $\times 0, 1, 2$:
larger ℓ_d , but the same total mass flux.

Plotted on linear axes, the cover decomposition $\alpha^p(l)$ peaks at an intermediate size in every case, as in observed cumulus fields (Plank 1969; Wielicki and Welch 1986). The reason is Eq. (10). Since $\alpha^p \propto l^2 \mathcal{N}$, a single power law would make either the smallest or the largest clouds dominate, depending on whether b is below or above -2 . The observed $b = -1.70$ below the break and $b < -2$ above it place the maximum at the break: knowing ℓ_d gives the dominant size, and vice versa. One-grid-box clouds cover somewhat too much area, probably a numerical effect.

The mass flux is area times velocity. Because the cloud-mean vertical velocity grows with cloud size (panel b), $\mu(l)$ peaks at larger sizes than α^p and is better defined. The smallest clouds carry close to nothing, apart from cloud base, where their large number somewhat compensates their low vertical velocities.

Robustness. With grids of 25, 50 and 100 m, the dominant size and its amplitude do not move, and neither does the largest cloud size; the cover of the smallest clouds tends to zero at finer resolution, so the intermediate peak becomes better defined. The total mass flux grows somewhat at finer resolution, which the authors attribute to poorer statistics of the largest clouds rather than to the smallest ones. A 3.2 km domain truncates the spectrum and moves the dominance to the largest size available. Shear tilts clouds and enlarges their projected area: the density broadens, the cover rises from 13 to 19%, and the break moves up, but the total mass flux stays about constant at 0.0145–0.0149 m/s, because the vertical mixing that destroys the conditional instability is unchanged. Brown (1999b) found shifts with resolution; the authors trace the difference to his method, cross-sections at one level rather than projected areas, and possibly to his Smagorinsky subgrid model.

The dominant size sits at the break, far above the grid scale, for cover and for transport

Conclusions

LES reproduces natural populations: a power law below the break, a distinct break above.

Support for the observed power laws and breaks; no evidence for exponential or lognormal forms.

A universal form, with the break ℓ_d as the only case-dependent parameter.

Break plus exponent give an intermediate dominant size, in the cover and in the mass flux.

The dominant clouds sit well above the grid scale; the smallest carry almost no transport.

Limits

One decade of scaling only; 1–2 m grids would be needed for another.

What sets the break is unexplained; the (sub)cloud-layer depth and the shear move it.

Too small a domain truncates the largest clouds.

Three shallow-cumulus cases, compared with published exponents, not the same scenes.

The disagreement with Brown (1999b) is not settled.

Snapshots only, so no cloud life cycle; no uncertainty quoted on b .

Realistic size statistics, one free length, and an unexplained scale break

§4a, §4f, §5. $b = -1.70$ below the break; the quoted exponents of Cahalan & Joseph (1989) and Benner & Curry (1998) come from remote sensing of real cloud fields.

Conclusions and limits

Conclusions of the paper. The shallow cumulus populations produced by LES match natural ones at several points: the size density is well described by a power law for the smaller clouds, with $b = -1.70$, and the scaling breaks down at a certain size above which the density falls off quickly. The resolution test suggests that the scaling continues below the typical grid spacing. These results support the power laws and scale breaks observed by Cahalan and Joseph (1989), Kuo et al. (1993) and Benner and Curry (1998), and give no evidence for the exponential form of Wielicki and Welch or the lognormal of Lopez. The authors call the power law a robust geometrical fingerprint whose physical explanation is an outstanding challenge.

The projected cover is uniquely determined by the size density, since it is just the density weighted by the projected area. The existence of the break, with the exponent above -2 below it and below -2 above it, causes a well-defined intermediate dominant size, in the cover and in the mass flux alike, consistent with observed cloud fields. Across the three cases the densities are remarkably uniform: the only variable is the position of the break, and with it the dominant size. Finally, the dominant clouds are not of resolution scale but significantly larger, which is fortunate, since the subgrid model governs the dynamics of the smallest clouds while the larger ones are better resolved; the smallest clouds contribute close to nothing to the vertical mass transport.

Limits. The power law is supported over about one decade, from the grid spacing to the break, and the authors note that 1–2 m grids would be needed for another decade. The mechanism of the break is the key open question: the depth of the (sub)cloud layer and the intensity of the shear are shown to move it, and subcloud humidity fluctuations are mentioned, but nothing is established. Too small a domain limits the largest clouds. There are only three shallow-cumulus cases, all shallow cumulus, and the comparison with nature goes through published exponents and dominant sizes rather than the same scenes. The disagreement with Brown (1999b) about the sensitivity to resolution is attributed to method and subgrid model, but not resolved. The statistics are built from instantaneous snapshots, so the study says nothing about how individual clouds grow and decay, and no uncertainty is quoted for the fitted exponent.